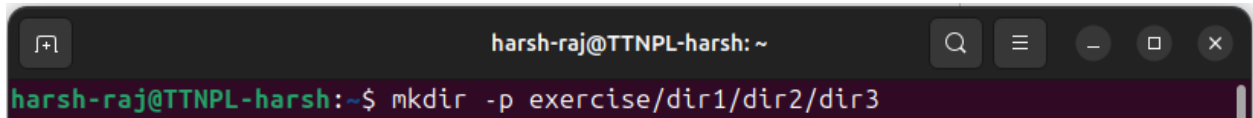


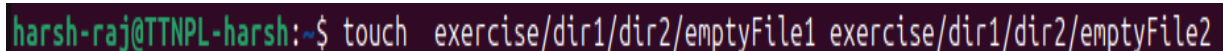
ASSIGNMENT 1

Q1) Create a directory "exercise" inside your home directory and create a nested (dir1/dir2/dir3) directory structure inside "exercise" with a single command.

A terminal window with a dark background. The title bar shows 'harsh-raj@TTNPL-harsh: ~'. The command 'mkdir -p exercise/dir1/dir2/dir3' is entered and executed, resulting in a green prompt and the command text.

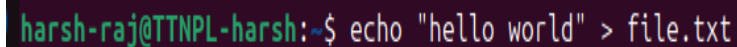
```
harsh-raj@TTNPL-harsh:~$ mkdir -p exercise/dir1/dir2/dir3
```

Q2) Create two empty files inside dir2 directory: emptyFile1, emptyFile2 in single command.

A terminal window showing the command 'touch exercise/dir1/dir2/emptyFile1 exercise/dir1/dir2/emptyFile2' being executed. The prompt is green and the command is white.

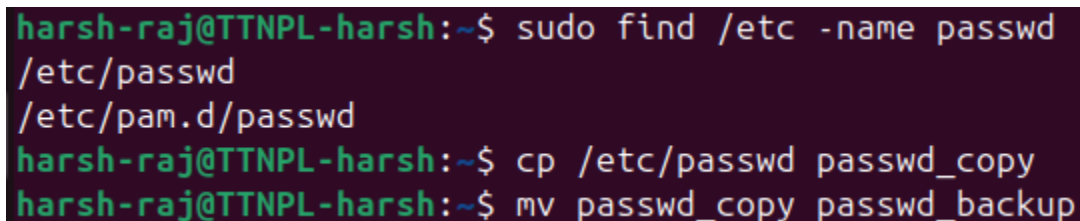
```
harsh-raj@TTNPL-harsh:~$ touch exercise/dir1/dir2/emptyFile1 exercise/dir1/dir2/emptyFile2
```

Q3) Create one file file.txt containing text "hello.world" and save it.

A terminal window showing the command 'echo "hello world" > file.txt' being executed. The prompt is green and the command is white.

```
harsh-raj@TTNPL-harsh:~$ echo "hello world" > file.txt
```

Q4) Find a "passwd" file using find command inside /etc. copy this file as passwd_copy and then rename this file as passwd_backup.

A terminal window showing three commands being executed: 'sudo find /etc -name passwd', 'cp /etc/passwd passwd_copy', and 'mv passwd_copy passwd_backup'. The prompts are green and the commands are white.

```
harsh-raj@TTNPL-harsh:~$ sudo find /etc -name passwd
/etc/passwd
/etc/pam.d/passwd
harsh-raj@TTNPL-harsh:~$ cp /etc/passwd passwd_copy
harsh-raj@TTNPL-harsh:~$ mv passwd_copy passwd_backup
```

Q5)Try reading passwd_backup file in multiple tools: less,more,cat,strings etc and find the difference in their usage.

CAT- It prints entire file at once, better for small files

```
harsh-raj@TTNPL-harsh:~$ cat passwd_backup
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
```

LESS- Allows scroll up/down, better for large files

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
:
```

MORE-Allows only forward scrolling, better for large files

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
--More-- (39%)
```

Strings- Shows only printable character,used for binary files

Q6)Find out the number of line in password_backup containing "/bin/false".

```
harsh-raj@TTNPL-harsh:~$ grep -c "/bin/false" passwd_backup
7
```

Q7)Get the first 5 lines of a file "password_backup" and Redirect the output of the above commands into file "output". Also, get the lines 6-10 from the above file.

```
harsh-raj@TTNPL-harsh:~$ head -n 5 passwd_backup > "output"
harsh-raj@TTNPL-harsh:~$ cat output
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

Q8)Create a "test" user,create its password and find out its uid and gid.

```
harsh-raj@TTNPL-harsh:~$ sudo useradd test
[sudo] password for harsh-raj:
harsh-raj@TTNPL-harsh:~$ sudo passwd test
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
harsh-raj@TTNPL-harsh:~$ id test
uid=1001(test) gid=1002(test) groups=1002(test)
```

Q9)Change the timestamp of emptyFile1,emptyFile2 which are exist in dir2

```
harsh-raj@TTNPL-harsh:~$ touch exercise/dir1/dir2/emptyFile1 exercise/dir1/dir2/emptyFile2
```

Q10) Login as test user and edit the "output" file created above. Since the permission wont allow you to save the changes. Configure such that test user can edit it.

a) Add group owner of the "output" file as the secondary group of testuser and check/change the "output" file permission if it is editable by group. Once done revert the changes

```
harsh-raj@TTNPL-harsh:~$ sudo usermod -aG harsh-raj test
harsh-raj@TTNPL-harsh:~$ chmod 770 output
harsh-raj@TTNPL-harsh:~$ ls
Desktop      exercise    newarchive  passwd_backup  snap          text.txt
Documents    file.txt    newarchive.tar  Pictures        Templates      Videos
Downloads    Music       output        Public          Test
harsh-raj@TTNPL-harsh:~$ ls -l output
-rwxrwx--- 1 harsh-raj harsh-raj 189 Jan 14 16:47 output
harsh-raj@TTNPL-harsh:~$ su - test
Password:
$ vi output
$ exit
```

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
hello this is me
```

```
"output" 6L, 206B
```

```
6,16
```

```
All
```

```
harsh-raj@TTNPL-harsh:~$ sudo chmod g-w output
[sudo] password for harsh-raj:
```

b) Make the file editable to the world so that test user can access it. Revert the changes after verification.

```
harsh-raj@TTNPL-harsh:~$ chmod o+w output
harsh-raj@TTNPL-harsh:~$ su - test
Password:
$ vi output
```

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
hello this is me
dghfgy
```

```
-- INSERT --
```

```
7,7
```

```
All
```

c) Change the ownership to edit the file.

```
harsh-raj@TTNPL-harsh:~$ su - test
Password:
$ echo "testing" >> output
$ exit
harsh-raj@TTNPL-harsh:~$ sudo chown harsh output
chown: invalid user: 'harsh'
harsh-raj@TTNPL-harsh:~$ sudo chown harsh-raj output
```

Q11) Create alias with your name so that it creates a file as "/tmp/aliastesting".

```
harsh-raj@TTNPL-harsh:~$ alias harsh='touch /tmp/aliastesting'
harsh-raj@TTNPL-harsh:~$ harsh
harsh-raj@TTNPL-harsh:~$ ls -l /tmp/aliastesting
-rw-rw-r-- 1 harsh-raj harsh-raj 0 Jan 14 22:48 /tmp/aliastesting
harsh-raj@TTNPL-harsh:~$
```

Q12) Edit ~/.bashrc file such that when you change to "test" user it should clear the screen and print "Welcome".

```
welcome
harsh-raj@TTNPL-harsh:~$
```

Q13)Install “zip” package.

```
harsh-raj@TTNPL-harsh:~$ sudo apt install zip
[sudo] password for harsh-raj:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
zip is already the newest version (3.0-13ubuntu0.2).
zip set to manually installed.
The following packages were automatically installed and are no longer required:
  libgl1-amd-gpu-dri libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
harsh-raj@TTNPL-harsh:~$ zip --version
Copyright (c) 1990-2008 Info-ZIP - Type 'zip -L' for software license.
This is Zip 3.0 (July 5th 2008), by Info-ZIP.
Currently maintained by E. Gordon. Please send bug reports to
the authors using the web page at www.info-zip.org; see README for details.

Latest sources and executables are at ftp://ftp.info-zip.org/pub/infozip,
as of above date; see http://www.info-zip.org/ for other sites.

Compiled with gcc 13.3.0 for Unix (Linux ELF).

Zip special compilation options:
  USE_EF_UT_TIME      (store Universal Time)
  BZIP2_SUPPORT       (bzip2 library version 1.0.8, 13-Jul-2019)
                      bzip2 code and library copyright (c) Julian R Seward
                      (See the bzip2 license for terms of use)
  SYMLINK_SUPPORT     (symbolic links supported)
  LARGE_FILE_SUPPORT  (can read and write large files on file system)
  ZIP64_SUPPORT       (use Zip64 to store large files in archives)
  UNICODE_SUPPORT     (store and read UTF-8 Unicode paths)
  STORE_UNIX_UIDS_GIDS (store UID/GID sizes/values using new extra field)
  UIDGID_NOT_16BIT    (old Unix 16-bit UID/GID extra field not used)
  [encryption, version 2.91 of 05 Jan 2007] (modified for Zip 3)

Encryption notice:
  The encryption code of this program is not copyrighted and is
  put in the public domain. It was originally written in Europe
  and, to the best of our knowledge, can be freely distributed
  in both source and object forms from any country, including
  the USA under License Exception TSU of the U.S. Export
  Administration Regulations (section 740.13(e)) of 6 June 2002.

Zip environment options:
  ZIP: [none]
  ZIP64: [none]
```

Q14)Compress "output" and "password_backup" files into a tar ball. List the files present inside the tar created.

```
harsh-raj@TTNPL-harsh:~$ tar -cvf newarchive.tar output passwd_backup
output
passwd_backup
harsh-raj@TTNPL-harsh:~$ tar -tvf newarchive.tar
-rw-rw-r-- harsh-raj/harsh-raj 189 2026-01-14 16:47 output
-rw-r--r-- harsh-raj/harsh-raj 2913 2026-01-14 15:58 passwd_backup
```


Q15) scp this file to test user

```
harsh-raj@TTNPL-harsh:~$ scp newarchive.tar test@localhost:/home/test
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:VOOW833KQUVn9GHfLj9LxlgzrRbS+TyQK2uA6mkxKCM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
test@localhost's password:
newarchive.tar
```

100% 10KB 14.3MB/s 00:00

Q16) Unzip this tar file by logging into the remote server

```
harsh-raj@TTNPL-harsh:~$ ssh test@localhost
test@localhost's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

29 updates can be applied immediately.
24 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jan 15 09:47:09 2026 from 127.0.0.1
$ tar -xvf newarchive.tar
output
passwd_backup
```

Q17)Download any image from web and move to desktop

```
harsh-raj@TTNPL-harsh:~$ curl https://cdn.britannica.com/16/234216-050-C66F8665/beagle-hound-dog.jpg > dog.jpg
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total  Spent    Left  Speed
100 159k 100 159k    0     0  935k      0 --:--:-- --:--:-- --:--:--  934k
harsh-raj@TTNPL-harsh:~$ mv dog.jpg Desktop
```

Q18)How to get help of commands usages.

```
CP(1)                                     User Commands                               CP(1)

NAME
    cp - copy files and directories

SYNOPSIS
    cp [OPTION]... [-T] SOURCE DEST
    cp [OPTION]... SOURCE... DIRECTORY
    cp [OPTION]... -t DIRECTORY SOURCE...

DESCRIPTION
    Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.

    Mandatory arguments to long options are mandatory for short options
    too.

    -a, --archive
        same as -dR --preserve=all

    --attributes-only
        don't copy the file data, just the attributes

    --backup[=CONTROL]
Manual page cp(1) line 1 (press h for help or q to quit)
```

```
MV(1)                                User Commands                                MV(1)

NAME
    mv - move (rename) files

SYNOPSIS
    mv [OPTION]... [-I] SOURCE DEST
    mv [OPTION]... SOURCE... DIRECTORY
    mv [OPTION]... -t DIRECTORY SOURCE...

DESCRIPTION
    Rename SOURCE to DEST, or move SOURCE(s) to DIRECTORY.

    Mandatory arguments to long options are mandatory for short options
    too.

    --backup[=CONTROL]
        make a backup of each existing destination file

    -b
        like --backup but does not accept an argument

    --debug
        explain how a file is copied. Implies -v

Manual page mv(1) line 1 (press h for help or q to quit)
```

We can get help of command using man command. By writing man and the command one wants to know about following man in the terminal we get the information about that particular command.

Q19) Create a symlink of /etc/services into /tmp/ports-info

```
harsh-raj@TTNPL-harsh:~$ ln -s /etc/services /tmp/ports-info
harsh-raj@TTNPL-harsh:~$ ls -l /tmp
total 56
lrwxrwxrwx 1 harsh-raj harsh-raj 13 Jan 15 10:17 ports-info -> /etc/services
```

Q20) You are appointed as a Software/DevOps Engineer in ABC media services. On your first day you need to troubleshoot a problem. There is a command “xyz” somewhere installed in that linux system. But as a new joinee you do not have any idea about where is that installed. How can you check that?

```
harsh-raj@TTNPL-harsh:~$ which pwd
/usr/bin/pwd
harsh-raj@TTNPL-harsh:~$ whereis ssh
ssh: /usr/bin/ssh /etc/ssh /usr/share/man/man1/ssh.1.gz
```